

Software Development Project



Official Team Name

CSE4104-7C-T03

Section

7C

Project Title

AI Study Material Generator

Team Information

Role	Name	Student ID
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Preliminary Project Description

The AI Study Material Generator is an intelligent educational platform designed to help students prepare study materials automatically using Artificial Intelligence. Users can upload lecture notes, PDFs, slides, or enter topics manually. The system will analyze the content and generate summaries, short notes, MCQs, flashcards, quiz questions, and important questions.

Why AI is Needed in This Project

- AI can summarize long educational content quickly.
- AI can generate MCQs and quizzes automatically.
- AI can identify important topics from study materials.
- NLP helps understand uploaded documents.
- AI can personalize learning content based on user needs.

Initial Technology Stack

Frontend

- HTML
- CSS
- JavaScript
- React.js

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Tools / APIs

- Gemini API / OpenAI API

Similar Existing Systems / Research

System Name	Features
ChatGPT	AI-based question answering and summaries
Google Gemini	AI educational assistance
Quizlet AI	Flashcards and quiz generation
Notion AI	Smart note generation

Research Sources:

- Google
- GitHub
- YouTube

Possible Features

- AI summary generator
- Smart MCQ generator
- Flashcard creator
- Quiz generation system
- PDF upload and analysis
- AI chatbot assistant
- Personalized study recommendations